MODULE 1

Exploratory Data Analysis for Finance

Descriptive statistics, distributions, and the detection of outliers.

Manaranjan Pradhan

www.aithinking.in

A Course on Quantitative Finance

CHAPTER I

Why start with ***exploratory*** analysis?

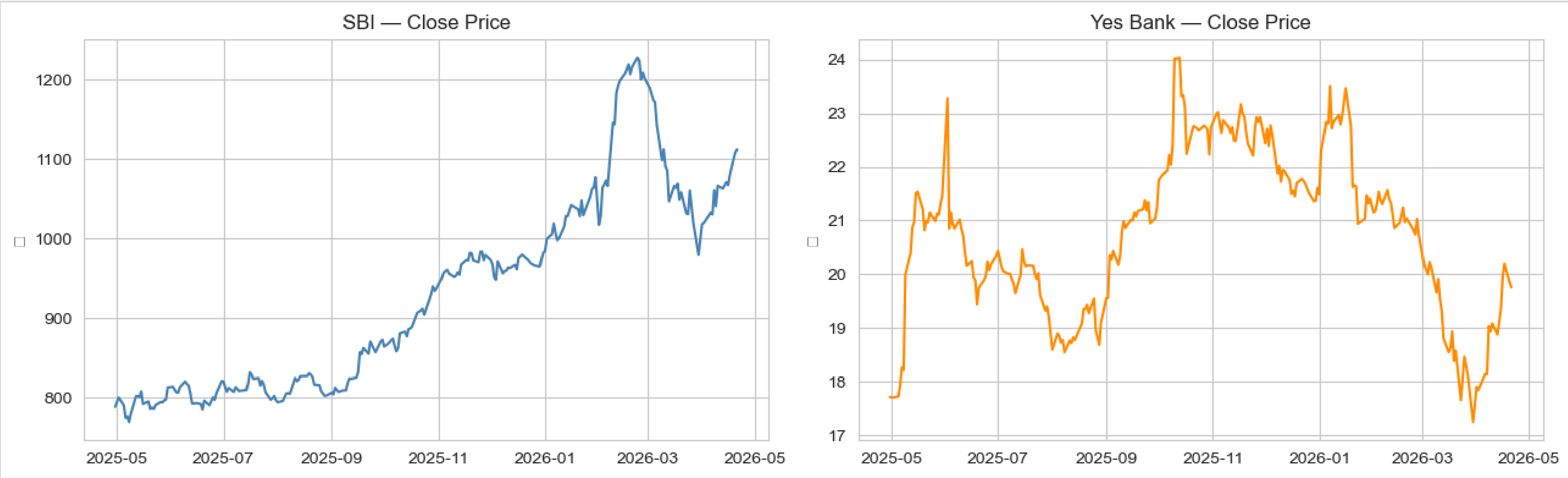
Before any model, we need a clear picture of two things: how much **return** each stock delivered, and how much **risk** — its *volatility* — came with it.

- What does the **return distribution** look like — centre, spread, shape?
- How **volatile** is each stock — what is its typical day-to-day risk?
- Are there **extreme observations** that will dominate later fits?
- How do **different stocks** compare on a common axis?

THE DATASET

Our two stocks

- **SBI** — large-cap public sector bank, ~₹1,100
- **YesBank** — low-priced private bank, ~₹20
- Price ratio **~55×** — same sector, very different scales



FROM LEVELS TO CHANGES

Prices drift. Returns don't.

Prices are non-stationary — they drift, trend, and live on wildly different scales across stocks. **Returns** are approximately stationary and directly comparable.

SIMPLE DAILY RETURN

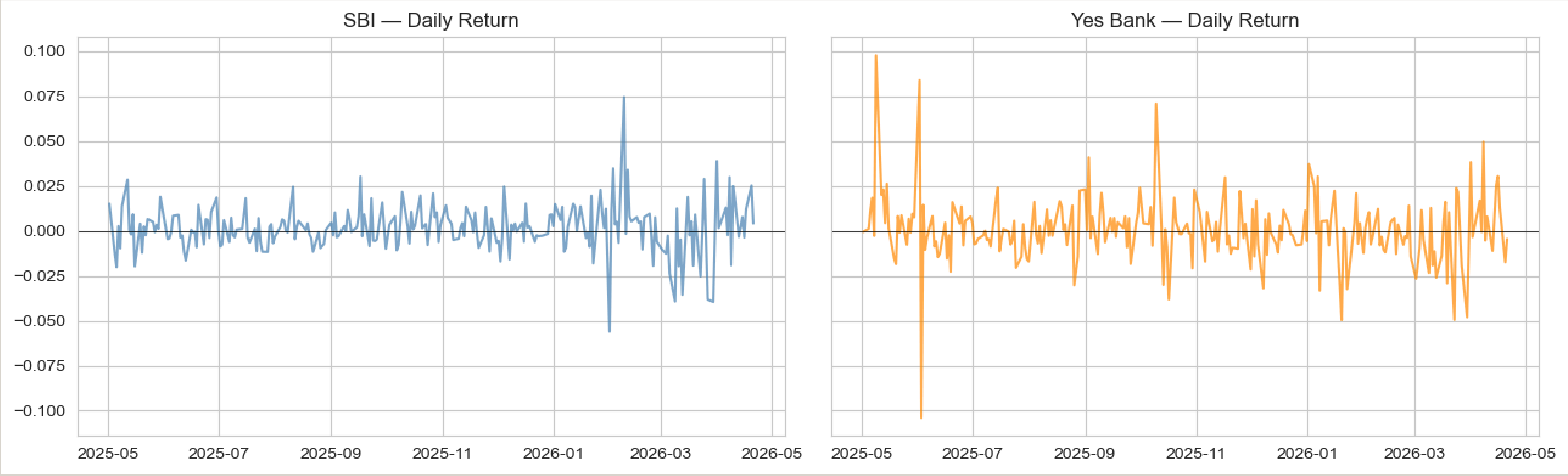
$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

For weekly returns, apply the same formula to a weekly price series (Friday closes).

FIRST LOOK AT RETURNS

A year of daily returns

- Most days hug **zero**
- Occasional **spikes** on both sides
- Turbulence arrives in **clusters**
- YesBank visibly **wider** — higher volatility



CHAPTER II · SUMMARY STATISTICS

The four moments of a distribution

Four numbers. Each answers a different question about the shape of the return distribution.

— **Mean** — where is the distribution centred?

— **Standard deviation** — how spread out? In finance this *is* risk, and we call it **volatility**.

— **Skewness** — is it symmetric, or is one tail longer than the other?

— **Excess kurtosis** — how *heavy* are the tails compared to a normal?

MOMENT 1

Sample mean

The arithmetic average return across the sample — the estimator of the true expected return.

SAMPLE MEAN

$$\bar{r} = \frac{1}{n}\sum_{i=1}^{n} r_i$$

For equity returns, the mean is typically close to zero — a small positive number reflecting the long-term equity risk premium, dwarfed by the noise around it.

MOMENT 2

Sample standard deviation

The typical size of deviation from the mean. In finance, this is our **canonical measure of risk** — we call it **volatility**.

SAMPLE STANDARD DEVIATION

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_i - \bar{r})^2}$$

The divisor n − 1 (rather than n) makes s^2 an unbiased estimator of the population variance.

Higher volatility → more uncertainty about tomorrow. *Investors demand higher expected returns as compensation for bearing more risk — the risk–return trade-off.*

MOMENT 3

Skewness — asymmetry

SAMPLE SKEWNESS

$$\text{skew} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{r_i - \bar{r}}{s}\right)^3$$

- **skew ≈ 0** — roughly symmetric distribution
- **skew < 0** — long *left* tail: crashes larger than rallies
- **skew > 0** — long *right* tail: upside surprises

Rule of thumb · industry — |skew| ≲ 0.5 *acceptable as approximately Gaussian* · **skew > 1** *(right-tailed)* → **log transformation** · **skew < −1** *(left-tailed)* → **power transformation**.
Equity returns are typically near zero or slightly negative.

MOMENT 4

Excess kurtosis — tail heaviness

SAMPLE EXCESS KURTOSIS

$$\text{excess kurt} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{r_i - \bar{r}}{s}\right)^4 - 3$$

The −3 makes the normal distribution's excess kurtosis equal to zero.

Rule of thumb · industry — |excess kurt| ≲ 1 *close to normal* · **> 1** → *fit a* **Student-t distribution.**
Equity returns almost always sit beyond +1 — tails are fatter than a normal would predict.

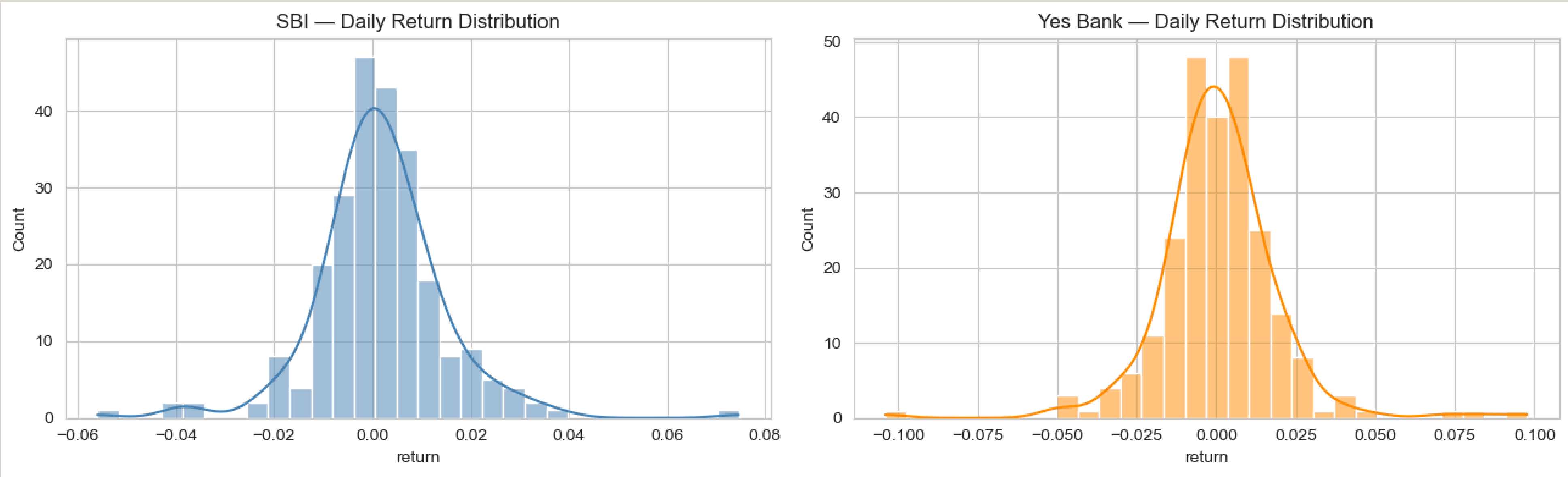
CHAPTER III

Visualising distributions — what the numbers hide

VISUALISATION · HISTOGRAM & KDE

The distribution, drawn

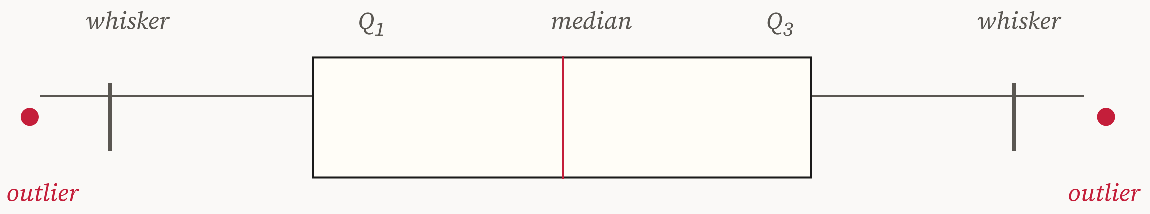
- **Histogram** — raw empirical frequencies
- **KDE curve** — smoothed density
- Mass clusters near zero · thin tails both ways



VISUALISATION 2

The boxplot, anatomised

A compact summary: five numbers in a single graphical mark.

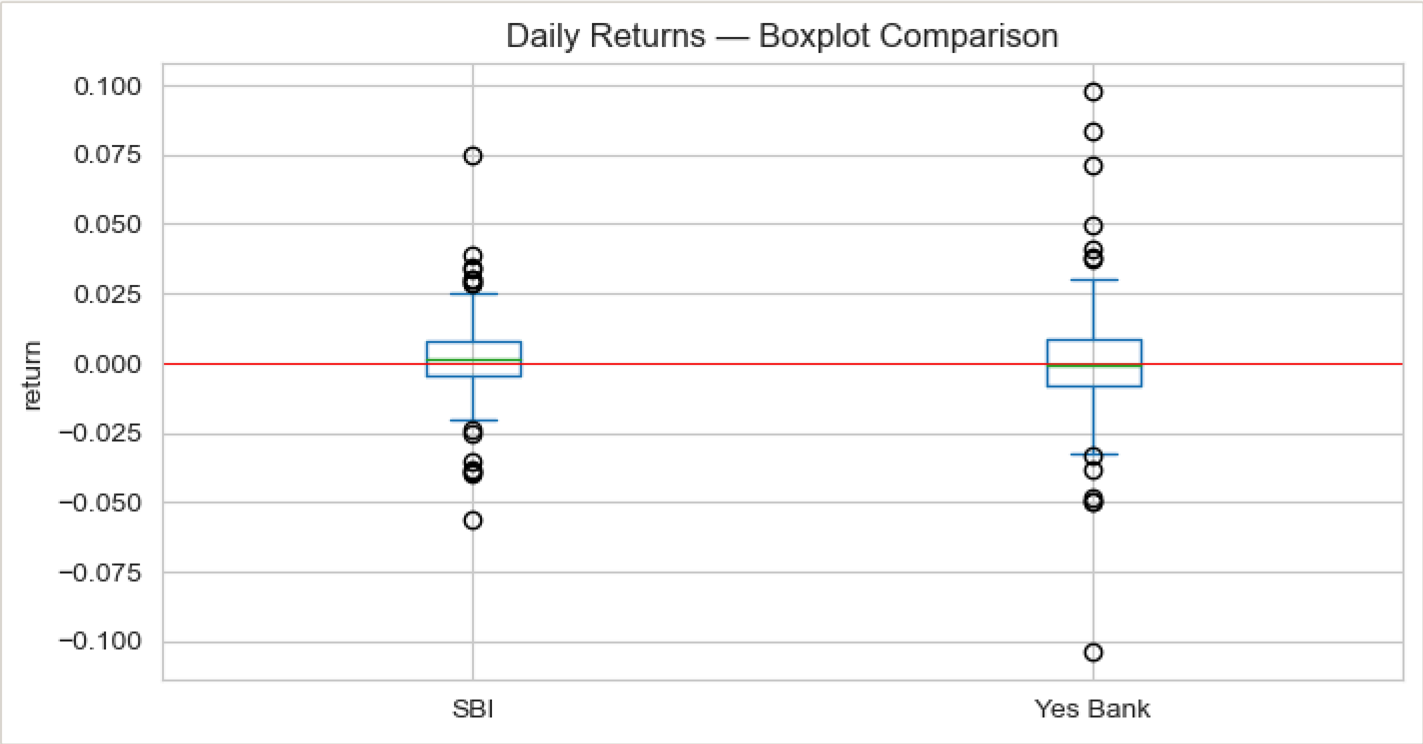


- **Box** spans the *IQR* (25th to 75th percentile) — the middle 50% of observations.
- **Whiskers** extend to the furthest point within 1.5×IQR of the box.
- **Dots** beyond the whiskers are statistical outliers (more on that next).

VISUALISATION · BOXPLOT

SBI versus YesBank at a glance

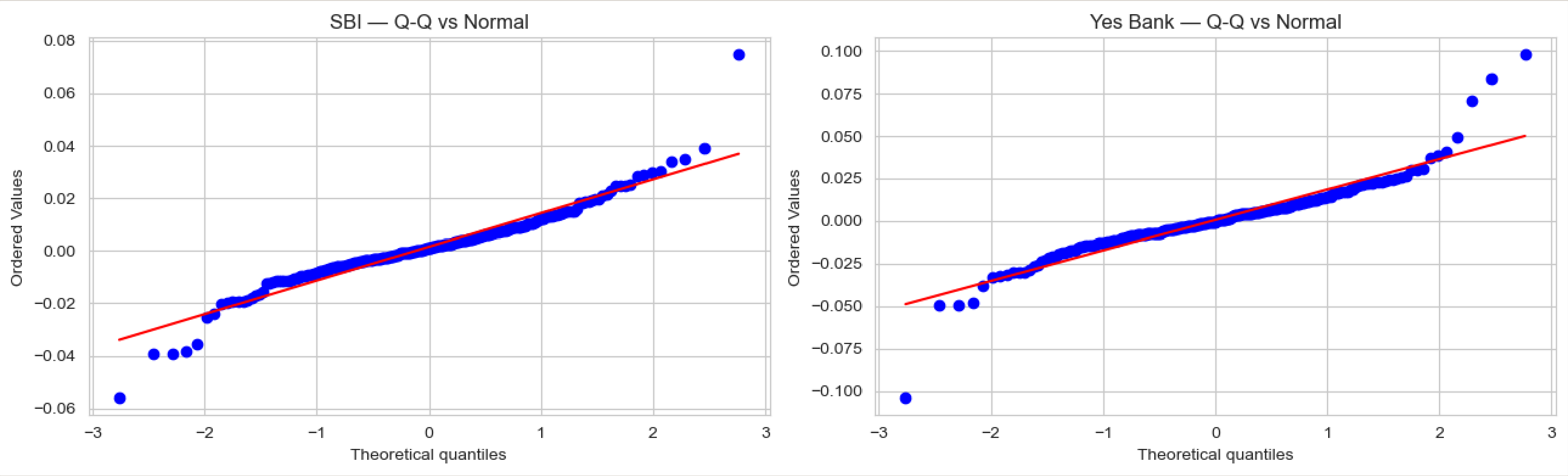
- YesBank's box is **wider** — more volatile
- More **outlier dots** beyond YesBank's whiskers
- Both medians near **zero** — roughly symmetric



VISUALISATION · Q-Q PLOT

Checking normality

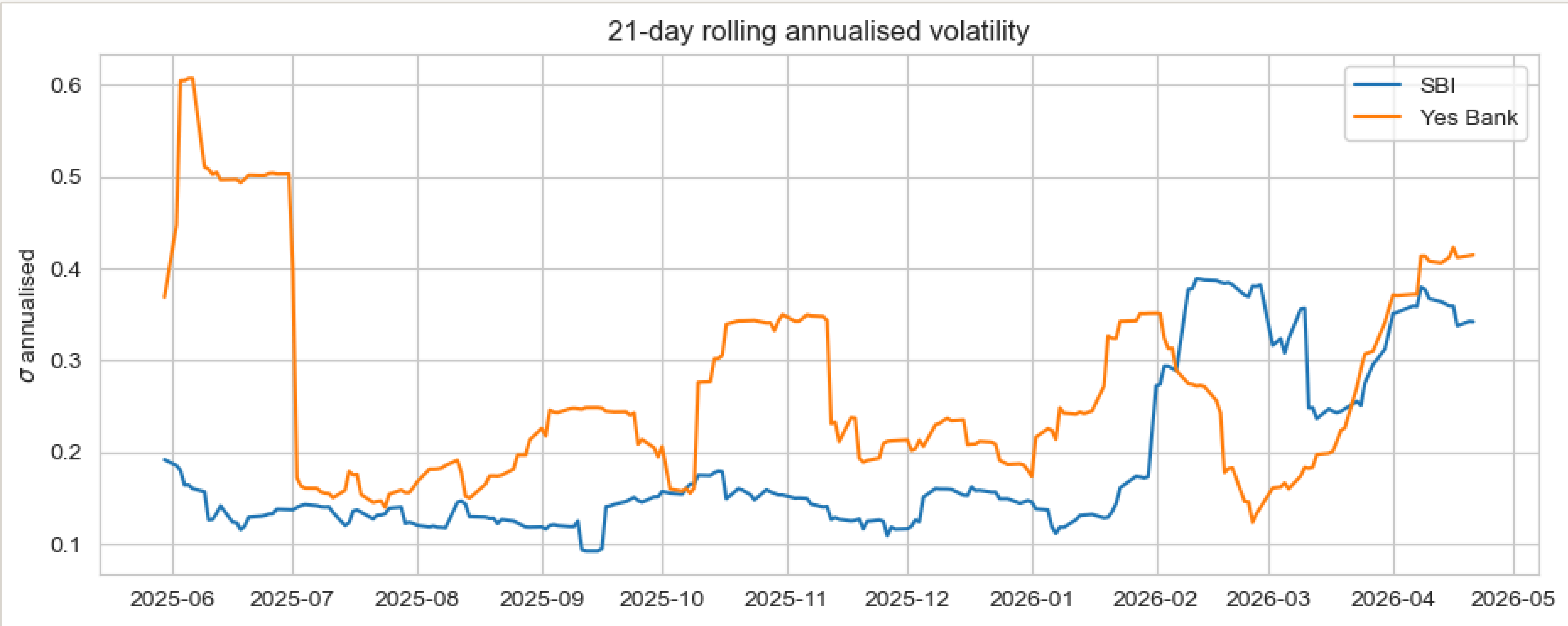
- Empirical vs. theoretical **normal quantiles**
- On the diagonal → **fits normal**
- Tails curving away → **fat tails**



STYLISED FACT

Volatility clusters

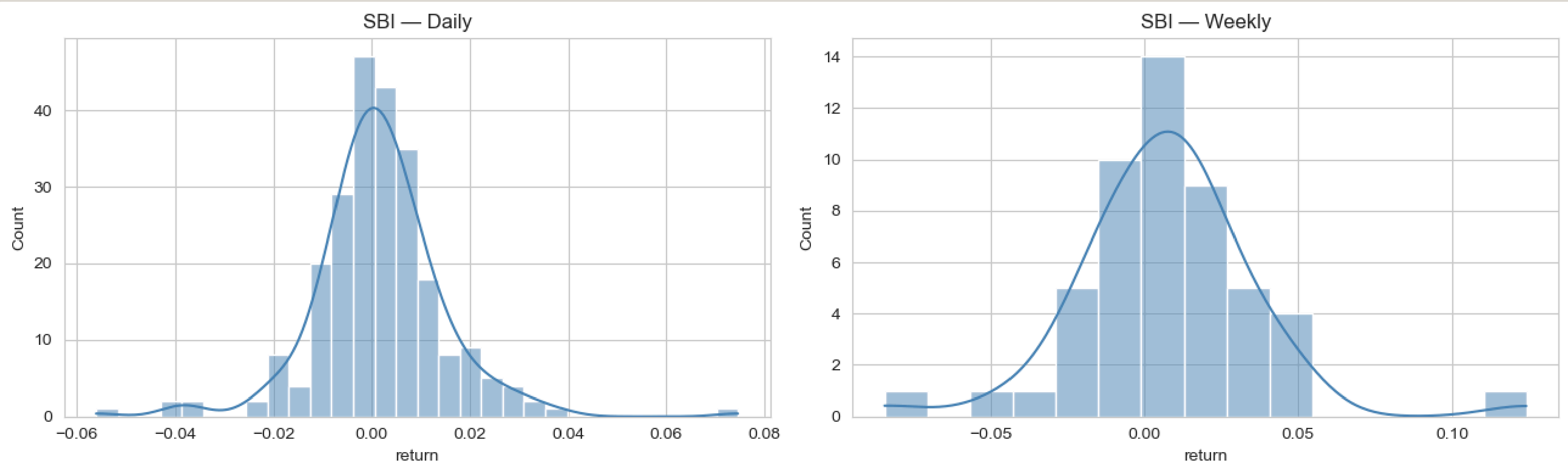
- High-volatility days **cluster together**
- Calm and turbulent periods **alternate** in regimes
- Documented since **Mandelbrot (1963)**



AGGREGATION

The Central Limit Theorem, in one picture

- Weekly = compounded **5 daily** returns
- CLT → sums tend toward a **normal**
- Weekly **less fat-tailed** than daily



CHAPTER IV

Outliers — two rules, two answers

RULE 1 · BOXPLOT-BASED

The IQR rule

An observation is flagged as an outlier if it falls outside the boxplot's whiskers — i.e. outside the *Tukey fences*.

IQR FENCES

$$\mathrm{IQR} = Q_3 - Q_1$$
$$\text{Flag if } \; r_i \; < \; Q_1 - 1.5 \cdot \mathrm{IQR} \;\; \text{or} \;\; r_i \; > \; Q_3 + 1.5 \cdot \mathrm{IQR}$$

Based entirely on **quartiles** — *robust* to a handful of extreme observations. A few outliers cannot themselves shift Q_1 or Q_3.

RULE 2 · STANDARD-DEVIATION-BASED

The three-sigma rule

3-SIGMA / Z-SCORE RULE

$$\text{Flag if } \quad |r_i - \bar{r}| \;>\; 3\,s, \qquad \text{i.e.} \qquad |z_i| > 3$$

Under a normal distribution, observations are distributed by the **68–95–99.7** rule:

Under normality, a |z| > 3 observation should occur about 3 days in 1,000. Under fat tails, it occurs *much* more often.

THE FAT-TAIL PARADOX

Why the two rules disagree

- The **3-sigma rule** uses the sample mean and standard deviation — both are *inflated* by the very outliers we want to detect.
- One extreme day enlarges *s*, which then **masks** other outliers.
- The **IQR rule** uses quartiles — which a few extremes cannot move.

> **Typical finding on equity data:** *the IQR rule flags more outliers than the 3-sigma rule. Neither is "right" — but each answers a different question, and a careful analyst reports both.*

CHAPTER V · COMPARING DISTRIBUTIONS

Standardisation — apples to apples

Stocks with different price and volatility scales can be put on a common axis by subtracting each series' own mean and dividing by its own standard deviation.

Z-SCORE / STANDARDISATION

$$z_t = \frac{r_t - \bar{r}}{s}$$

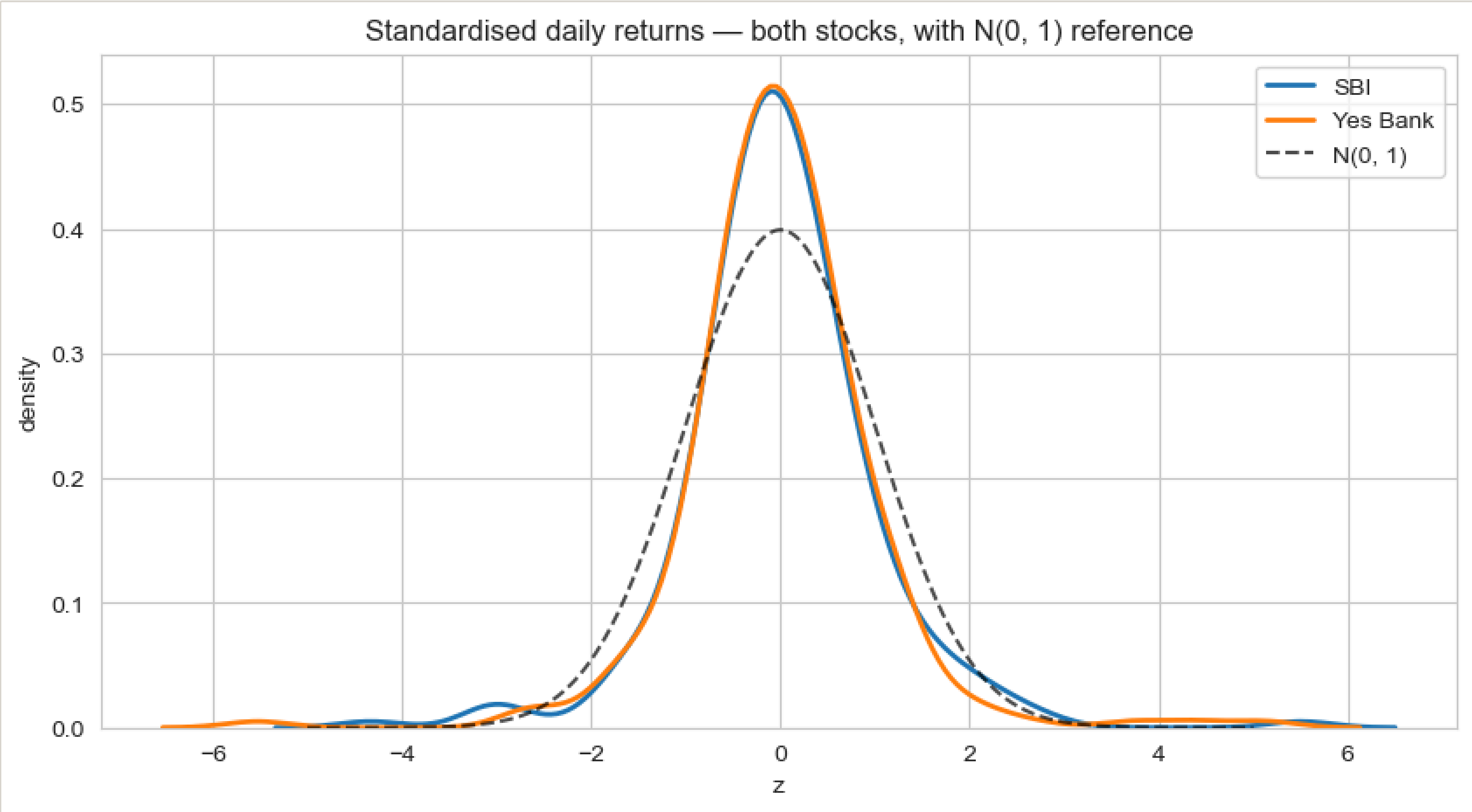
Every standardised series has mean 0 and standard deviation 1 by construction.

What remains after standardisation is **shape** — skew, kurtosis, the texture of the tails.

ON A COMMON AXIS

Two stocks, one scale

- Both centred on **0** with unit std
- Dashed **N(0,1)** — the reference curve
- Remaining bulges → genuine **shape** differences



CLOSING

Five things to remember

- Study **returns**, not prices.
- Equity returns are **fat-tailed** — the normal distribution under-predicts extremes.
- Volatility **clusters** — turbulent days arrive in bursts.
- Outliers: the **IQR rule** is robust; the **3-sigma rule** is sensitive. They disagree for a reason.
- **Standardisation** lets different stocks share an axis — and only then are shapes comparable.

> *Visualisations reveal what summary statistics summarise.*
>
> *Use both.*